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Intelligent Learning Systems for Inclusive Education in Ghana: Towards an Effective Engagement with Hard of Hearing Students

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***Abstract*—This research attempts to address some challenges faced when facilitating an inclusive classroom experience for Deaf/Hard of Hearing students (D/HH) in Ghanaian schools and universities by proposing an intelligent learning system. Hence, this research contributes an Intelligent and Inclusive Learning System (IILS) with two main components: signWithMe subsystem and audio-visual transcription system (AVTS) subsystem.**

The signWithMe subsystem presents a Ghanaian sign language learning management system (LMS) that is motivated by the limited sign language literacy in Ghana. This subsystem consists of a sign language dictionary, an E-resource, an E-forum, and an E-payment system. It supports inclusive learning of the Ghanaian sign language for both Deaf and Non-Deaf students to interact with the system. This ensures improved diversity and inclusion in our Ghanaian society. Also, the AVTS focuses on improving the teaching and learning experience of Deaf students in Ghanaian schools. The AVTS contributes a novel artificial intelligence approach that uses a counter-checking of Google speech-to-text and lip-reading-to-text transcriptions of the Ghanaian sign language in inclusive university lectures. These two subsystems provide services in line with the mandate of the tenth and fourth Sustainable Development Goal (S.D.G.) of "*Reduced Inequality*" and "*Quality Education*" respectively.

Finally, the IILS was analysed per the two main components (i.e., signWithMe and AVTS) and are considered to have high level of *usability* by *20 Deaf* and *5 Non-Deaf* users.

***Index Terms*—Inclusive education, Deaf/Hard of Hearing, Sign Language, E-learning system, Text-to-speech, Transcription, Translation**

I. Introduction

A. Background

According to the World Health Organisation [1], over 1 billion people live with some form of disability. Out of that populace, about 430 million people, including children, have complications with their hearing ability [2]. The World Health Organisation defines hearing loss as someone who cannot hear within the hearing thresholds of 20dB or better in both ears. Some of the causes of hearing loss and Deafness include genetic factors, childhood diseases and infections [2]. Hearing loss conditions result in one becoming hearing-impaired, Deaf or hard-of-hearing (D/HH). In Sub-Saharan Africa, there are about 8.9 million [3] Deaf children. The census report from the Ghana Statistical Service showed that as of 2010 [4] in Ghana, with a population of 24,658,823 people, 737,743 were hearing impaired. Most African societies stigmatise the Deaf and refer to them as "deaf and dumb" [5]. However, some literature [6] indicates that assistive technologies can help students with disabilities to enhance and improve their independence in academic and employment tasks. However, such technologies could vary significantly in cost from one device to another and sometimes can be too expensive for families or schools to afford. Hence, the speech recognition-based operation of devices such as prevalent smartphones should be leveraged on for accessibility and inclusion in our schools and communities [7]. Also, some of the students in one of the Ghanaian public universities, complained about the interpreting style when a "new" sign language interpreter takes over from an "old" sign language interpreter [8]. This reveals the need for government bodies to standardize the Ghanaian Sign Language at all levels of educational institutions to avoid confusing the D/HH students.

There is a communication gap between sign language and oral language [9], [10], and members of society must support families with Deaf children by learning how to communicate with the Deaf children to promote inclusion, and diversity in the society. Furthermore, the illiteracy rates among the Deaf in most countries are above 75 per cent [11], and very few get into higher education. As stated in [12], Deaf children who learn sign language during the childhood stage or school-going period have higher analogical reasoning skills than those who acquired sign language at an adult stage. Hence, there is a need to educate both the Deaf and hearing on sign language to ensure smooth communication and hence, harness the growth of the Deaf community. In addition, main stream educational institutions should make provisions for inclusion of at least their hard of hearing students.

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B. Research Aim, Objectives and Question

This research aims to first provide educational software that will help the teaching and learning of Sign Language and bridge the communication gap between the hearing and the Deaf students. This would be achieved by developing signWithMe system that is composed of a sign language dictionary, a translator, an e-resource, an e-forum and an e-payment system. The second aim of the IILS is to improve the speech-to-text and lip-reading-to-text transcriptions and translations in inclusive education environments with the audio visual transcription system (AVTS). To achieve the stated aims above, five objectives are provided below:

1) *OB1*: Development of a Sign Language Dictionary which consists of videos of the sign language alphabet, numbers, days of the week, and a few phrases.
2) *OB2*: Development of a Sign Language Translator for the Ghanaian sign language.
3) *OB3*: Development of a Sign Language E-payment System which enables the purchase documents for their learning.
4) *OB4*: Development of a Sign Language E-Forum which enables users to interact and share information about the Deaf and Sign Language.
5) *OB5*: Build a system that incorporates existing technologies that supports the classroom experience of D/HH students, i.e., to access course materials, participate in class discussions, and attend an in-person class.

The aim and objectives outlined above lead to the research question: "*How can an inclusive e-learning system improve classroom engagement for D/HH students in Ghanaian educational institutions?*" This research is motivated by the Sustainable Development Goals 10 (Reduced Inequality) and 4 (Quality Education) [13].

II. Literature Review

Assistive technologies enable individuals with disabilities to perform functions that might otherwise be difficult or impossible [6]. Hearing assistive technology helps individuals with hearing needs in listening, comprehending, or recognising sounds and enhancing sound frequencies, making them much clearer [14]. Some researchers explained that Deaf students fail to express their feelings and views to the hearing due to the communication barrier [8]. We discuss the related work in the following groupings: Deep learning and computer vision, OCR Web applications for inclusion, Speech-to-Text applications, and Mobile/Web E-Learning System.

A. Deep Learning and Computer Vision

In five peer-reviewed papers from IEEE, three researchers [8], [15], [16] used deep learning and computer vision in developing sign language systems.

Moreover, researchers in some of the reviewed papers included Support Vector Machine [8], [15] in their implementation. Although, it could be expensive and lack processing power. Their research aim was to focus on recognising the fingerspelling of the manual alphabet and manual numeric digits. They performed this research using three different classification models. Thus, the log-sigmoid neural network, Elliot neural network and the SVM.

B. OCR Web Applications for Inclusion

Other researchers from the reviewed papers [15] used OCR and Web 2.0 features that supported several sign languages. OCR makes use of optical character recognition. The process involves converting the scanned image into a binary image and then doing a connected component analysis for extracting character outlines. A case study in Japan demonstrated how a centralised-based learning e-system increased interactivity between the teachers and D/HI students [17].

C. Speech-to-text Applications

An inclusive classroom requires the use of real-time captioning using a Computer Assisted Real-time Translation (CART) captioner to support D/HH. A captioner (either a person or through a system) is usually present during class and produces an exact transcript of everything heard in the classroom in real-time [18], [19], [20]. CART requires the use of a good microphone in the classroom so that the captioner can listen to the lecturer's voice and transcribe accurately. However, a core limitation to the development of a captioning system is in its ability to recognise and transcribe multiple voices in a classroom discussion where the students do not have individual microphones [18].

In spite of the cutting-edge nature of AI today, the average word-error-rate (WER) for speech-to-text is far from 100%. As per benchmarks published in March 2020 [21], Amazon had an accuracy of 73% (i.e., 27% WER), Microsoft was 78% accurate, Google came in at 79%, and Rev.ai (a dedicated speech-to-text engine provider) scored a slightly better 84%. That said, Google Speech API has the largest text database.

D. Mobile/Web E-Learning System

Researchers have proved that E-learning systems in the form of mobile and web platforms can support the classroom experience of D/HH. An E-learning system was implemented in Malaysia for D/HH between 7 to 8 years old who wanted to learn alphabets and words using text and sign language [22]. The researcher designed the system using storyboards to describe the customer journey when using the system. They incorporated various sections to help guide the system's users and measured it with different HCI system principles such as consistent colour-coding, accuracy, and performance.

Based on the insight obtained from the related works above, a web application that can also be accessed on mobile phones would be implemented to make it convenient and accessible. Here, some technologies like HTML, CSS, JavaScript, Ajax, MySQL, PHP and frameworks like Bootstrap will be used in developing the web application system.

III. METHODOLOGY

The methodological approaches used include:

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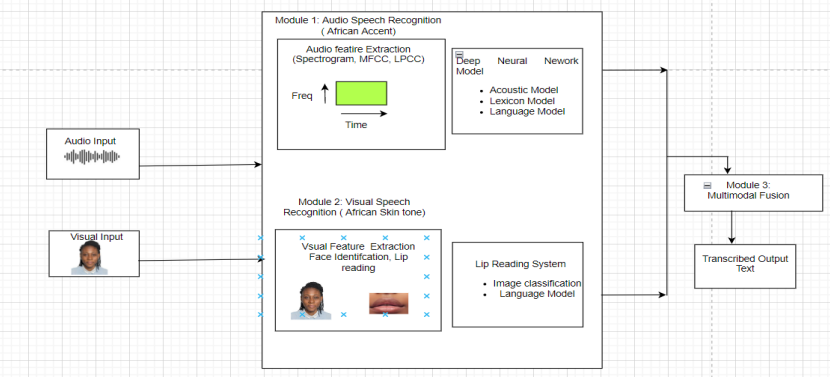


Fig. 1. Audio-Visual Transcription Methodology

A. Consent of Participants

The consent requirement of the Institutional Review Board (IRB) of Ashesi University is essential for this research in order to meet the standards of conducting ethical research which respects the privacy of the disabled. This is in accordance with the international standards for the protection of human participants in academic research. The IRB clearance forms obtained were distributed and read to the participants before engaging with them.

B. Data Gathering

1) Online Survey: An online survey was created and distributed to specific stakeholders and disability organisations in Ghana. The survey was created to gain more information from PWD on some assistive technologies they use to carry out general and education-specific activities. The survey also targeted school administrators to find out how they promote an inclusive education of PWD and non-disabled persons in their educational institutions.

2) Focus Group Interviews: In addition to the online survey, several focus group sessions were conducted to get qualitative data on the experiences of persons with disabilities. First the interviewees were asked for their consent and assured of privacy of the information shared during the focus group session. Thereafter, interviewees were asked questions and they gave their responses which were recorded and translated into themes.

C. System Performance and Usability Metrics

This is to measure how efficiently a D/HH student can perform tasks on the system through the system functionalities. It includes aspects such as *task response time* which measures how long a D/HH can complete a task efficiently. Also, the system will be evaluated using Human-Computer Interaction (HCI) principles such as simplicity, consistent colour schemes, and free from errors.

D. Audio Visual Transcription Methodology

The Audio Visual Transcription system captures both the audio and visuals of lips of the instructor during a lecture (refer to Figure 1). The audio input undergoes audio feature extraction (based on the Google Speech Translation model), in a component called Audio Speech Recognition with particular emphasis on African accents. Also, the visual input undergoes visual lip feature extraction (based on the RNN and LSTM model [23]) in the Visual Speech Recognition component with particular focus on African skin tones. Then, the audio and visual features are synchronised in the Multimodal Fusion component. Finally, a text transcript is generated.

IV. Data Collection and System Design

The duration of the data collection process spanned a period of two months. Additionally, one-on-one interviews and focus group conversations with 106 Deaf persons were conducted with D/HH students and other stakeholders in the disability space in Ghana. The contact institutions for the primary research include The Special Needs Department at the University of Ghana Legon, Deaf Congregation of Church of Christ, Circle and Demonstration School of the Deaf & Blind, Mampong, Eastern Region. At University of Ghana's Special Needs Department, the focus group involved 5 Deaf students, 1 Special Needs Assistant and the Head of the department, and the main insights were that Deaf students depend on mobile applications, are encouraged to sit in front-rows during lectures, and use of manual methods. Also, at Church of Christ we interacted with 79 congregants where we noted that the Deaf rely entirely on sign language in communication and are not conversant with Speech-to-Text applications, and they feel safe and accepted in the church community due to measures instituted to accommodate them and their special needs. Finally, at School of the Blind, Mampong, we interacted with 20 Deaf students where we noted that brilliant students in Maths but struggle with English courses, and students engage in informal jobs to earn a living.

1) Data Sampling: A breakdown of the total sample size involved in this research is as follows: The online questionnaire had 21 Responses. Many Deaf students depend on Mobile applications since it is easier and more convenient to use. Most of them use other technologies such as speech-to-text applications during classes and are highly encouraged to sit in the front rows of the class. The participants rely entirely on sign language in communication and it was also noted that many Deaf in Ghana are not given a chance to pursue science-

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related courses at the university level and are regularly pushed to social science courses.

First, on the first home page, the user logs in and gains the opportunity to utilise all the functionalities of the system. The user lands on the second home page and can navigate the about page, the dictionary, the translator, the e-resource, the e-forum and the e-payment system functionalities. After which, the user can log out of the system.

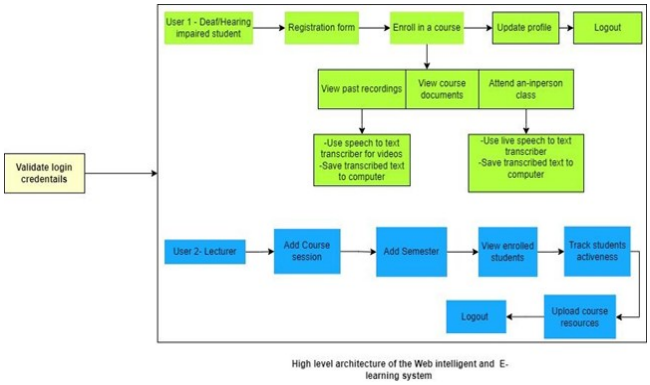


Fig. 2. Interactions with the AVTS

The insights derived from the primary research informed the design of the system to support the educational needs of the D/HH in a Ghanaian tertiary institution (refer to Figure 2). Some of the suggested features given by the participants of the focus group session and online survey include:

- The AVTS should have a speech-to-text application to help the Deaf students read the transcripts of the lectures and comments or questions from colleagues (i.e., Deaf and Non-Deaf),
- The text should be magnified to 18 pts,
- The system should be laptop and mobile compatible because most D/HH students mostly use mobile applications.

Based on the features mentioned, the proposed system is an Intelligent and E-learning web application that allows a D/HH:

- Enroll in a course
- Access course materials
- Attend an in-person class, or view course recordings with the help of a speech to text transcriber that takes the lecturer's audio or audio from a video and translates it to text.
- Then, the transcribed text is displayed on the student screen, which can be saved in any known document extension such as .pdf, and .doc format. On the other end of the system, the lecturer can login as an Administrator to monitor the student's login activities and upload course materials.

Based on both the system and user requirements provided above, the IILS composed of the signWithMe subsystem and AVTS is designed as a web application system (and to be implemented using HTML, CSS, Javascript and Bootstrap) to ensure user-friendliness. The IILS has been designed with clear and bold writing, warm colours, and features that ensure the signWithMe system is highly interactive and accessible.

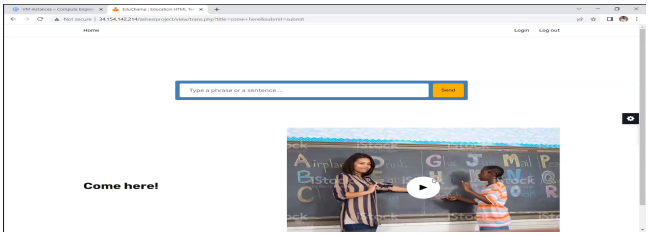


Fig. 3. signWithMe Sign Language Translation Video Search

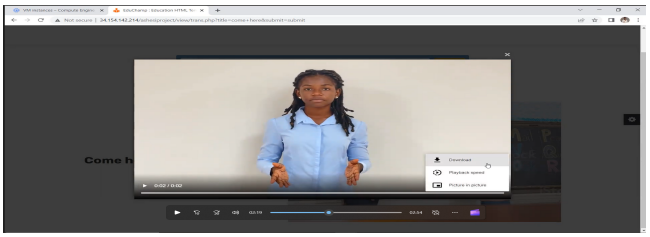


Fig. 4. signWithMe Sign Language Translation Video

A. signWithMe System Design Architecture

The signWithMe system is built as a web system with an efficient backend architecture that interacts with the database and updates the frontend of the system.

1) Web Application Architecture: The MVC approach will be used in implementing the signWithMe system. The system will be structured into three logical components, Model, View and Controller. When users query the signWithMe system for videos on the alphabet, the controller connects with the model—the model also connects with the videos' database. The database has the videos in the alphabet. Then, the view displays the video on the alphabet for the user to learn and interact with the system.

2) Database schema: The database has been designed to contain the entities and attributes of the data. The database schema contains the relationships between tables in the database and their data types.

V. AVTS Features

On the student part of the website, the Speech-to-text feature is the core functionality to support Deaf/Hard of Hearing students in a live class and this was implemented using two approaches. The first approach was developed using Google cloud speech API, and the transcriber function built in Python. The second approach was developed using Web Speech API in JavaScript, which interfaces easily with Google Chrome browser.

The transcription code developed is written in Python using several modules and libraries from Google Cloud API and native Python packages imported into a Python script. It takes streams of inputs by accessing the microphone of the user and stores the audio streams temporarily in a buffer. The audio stream input captured is sent to Google Cloud Speech API, where it is converted from speech to text in real time. To counter-check this translation, the camera module of a laptop is interfaced with a lip-reading module based on Long Short-

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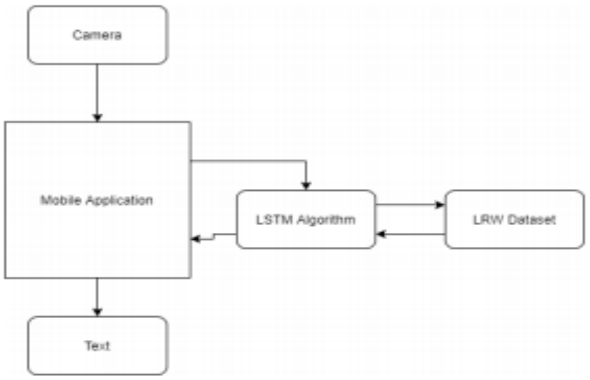


Fig. 5. Lip-Reading LSTM Architecture

Term Memory (LSTM) algorithm (see Figure 5). LSTM is an artificial recurrent neural network (RNN) architecture.

After transcription is done, it displays the transcribed text on the terminal of the users' machine. The transcribed text is displayed on a web page using Python's FastAPI which supports web applications. The user is able to view the transcribed text through a specific route which in this application is the '/speak' route

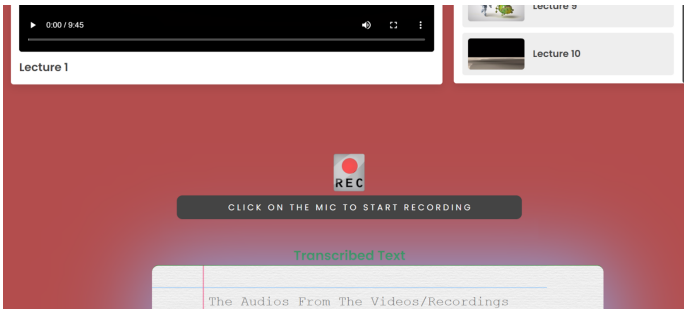


Fig. 6. AVTS Transcription Page

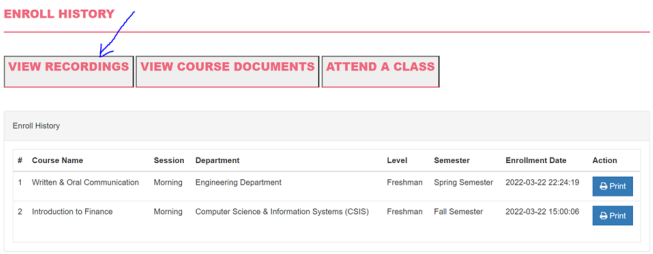


Fig. 7. AVTS Audio Recording List Page

The screenshot of the final transcription system (i.e., AVTS) is provided in Figure 7.

VI. Discussion of Results

A. Audio Visual Translation System

The translation functionality of the system was tested and measured over three criteria which are: accuracy, performance, and usability. Additionally, other features of the system for both the lecturer and the student were tested and evaluated using human computer interaction (HCI) system design principles such as efficiency, consistency, and error free.

B. Word Error Rate

The word error rate is is a common metric used to compare the accuracy of the transcripts produced by speech recognition and transcription service. The AVTS system built used two transcription services to measure the Word Error Rate of translating audio recordings. After series of tests with various audio recordings, an average percentage value was derived relative to their word-error-rate (WER) as shown in the Table I below:

TABLE I
WER of Speech Translation Systems

Speech Models α	**WER (%)** β	**WER with Accent (%)** γ
Web Speech API	65	76
Google Cloud speech API	25	45

During a demo session, the Web Speech API and Google Cloud speech API transcription functionality were tested using several 100-words scripts read out for transcription. The accuracy of the transcribed text differed between the two speech-to-text API's. The accuracy was measured by comparing the text from the original script with the words captured by each of the API's. After each transcription, the following were recorded: the number of correct words, the number of incorrect words captured, and the time spent in a transcription. In each of the 100-words scripts used, both APIs performed well in transcription accuracy. However, the Google Cloud speech API had the highest rate of accurate words because it supported wider range of languages such as EN-GH (Ghanaian-English) and vocabularies than the Web Speech API. Next, the performance of both APIs was extremely high and effective during translation in real time. However, both transcription services depend on strong internet connectivity and there are lags in transcription when there is unstable network.

C. User Testing of the AVTS

The testing activity of the E-learning system was carried at in the School of the Blind Deaf Mampong. The school for testing was selected because it had a larger population of D/HH students in the upper classes of their high school journey which is also pre-university levels. During testing, the time given to interact with the 20 students was limited because the students had classes in session. The D/HH students had a positive experience with the system's features and suggested that the audio from the recorded videos on the system should not be too fast so that they can follow while the transcription is ongoing.

However, ten D/HH students were selected at random to interact with the student part of the learning system and one teacher was also selected to interact with Lecturer section of the system.Each student interacted with the different functionalities of the system for 10 minutes and after which provided

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their feedback. The primary language used for communication in the School of the Deaf and Blind is sign language. Thus, through the help of a sign language interpreter, the system was described to the selected students, and they were encouraged to explore the features of the system and provide feedback on their experience.

D. Testing of signWithMe System

For the signWithMe subsystem, there are two tests implemented, the unit testing of all relevant methods and functions of the system and the user testing, which was conducted in the Mampong School for the Deaf. Through this process, the signWithMe subsystem has a sign language dictionary, translator, e-forum, e-resources and e-payment subsystems. The application system also meets the non-functional requirements.

Thus,

- the log-in/sign-up system ensures security by ensuring users log-in to access the functionalities and features of the application system. The system verifies the user's email address before he/ she can access the application's content.
- the Graphic User Interface (GUI) of the application system has been designed to ensure user-friendliness and interactivity. The navigation of the site is manageable and comprehensible, and
- the database and the hosting site have been secured such that the application system would always be available for users.

The signWithMe application system was evaluated by ten Deaf students, two teachers at the Mampong School for the Deaf, and some students at Ashesi University. They gave important feedback that the application is easy to use and navigate. One Deaf student, shared that because the videos are short and demonstrate a particular sign, it makes it comprehensible. Another Deaf student, shared that the system had bold and clear writings, making reading easier. The students were pleased with the explanation given on each of the signed words as it made learning more accessible. Also, the Deaf students looked through the signWithMe sign language dictionary and gave feedback on how some words should be signed. They also shared that the system can be a mobile application to be accessed on the phone.

VII. CONCLUSION AND LIMITATIONS

This research explored what accessibility technologies and inclusion concepts will improve the classroom engagement for a D/HH student in Ghanaian schools. Through literature review and stakeholder engagements in the disability sector, specific system features were highlighted. As a result, the IILS was developed to provide a web platform where people can learn and practise sign language, and D/HH students to benefit from audio-visual transcription in their classrooms. Additionally, the IILS developed supports both students and lecturer. It can be concluded that since objectives 1-5 were achieved, the IILS reasonably improve classroom engagement for D/HH students in Ghanaian educational institutions. This research also focused on the Sustainable Development Goals 10 (Reduced Inequality) and 4 (Quality Education).

The IILS can be improved especially with the Audio-Visual Transcription System (AVTS). During transcription, there were instances where the speech-to-text APIs displayed wrong words due to a noisy environment, an inaudible voice or the teacher speaking fast. The Google Speech API could not accurately identify words pronounced with African accents (45% WER). Here, more machine learning optimisation approaches like language and hyper-parameter models can be explored to improve the efficiency of the Google speech-to-text transcription. Also, it must be trained using data sets with audio recordings from Ghanaians to help in producing better results for the speech-to-text transcription.

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